

PVTRACE · PROBLEM STATEMENT

PVTrace problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Problem with cited evidence, users, context, constraints, prior work and co-design checklist for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

PVTrace problem statement

Installers and reuse programs judge solar modules by eye or by a single open-circuit voltage reading, which misses the faults an IV curve shows at once: a shaded or cracked substring, a failed bypass diode, high series resistance from corroded connections, or plain power loss. The instrument that shows these, a field IV curve tracer, is priced for utility-scale commissioning crews, so small installers, repair shops and second-life panel programs go without. A single-module commercial kit such as the Seaward PV210 lists at \$3,799 ([Transcat](#)).

■ Why it matters now

- **Volume.** Global PV capacity rose to nearly 3 TW at the end of 2025, with about 698 GW installed in that year alone ([IEA PVPS Snapshot 2026](#)). Panel waste could reach 78 million tonnes by 2050 ([IRENA and IEA PVPS, 2016](#)).
- **Early removal.** About 70 % of modules are decommissioned before the end of their 25 to 30 year design life ([Huang and Long, *Nature Communications*, 2026](#)), so many removed modules still work.
- **Slow degradation.** Across nearly 2,000 field measurements, the median power loss is about 0.5 % per year ([Jordan and Kurtz, 2013](#)); a 10-year-old module in good condition typically keeps most of its rated power.
- **Unknown quality in second-hand trade.** Refurbished modules exported to secondary markets show 10 to 20 % power degradation and inconsistent quality, and most of the projected PV waste in Africa and the Middle East will come from imports ([Huang and Long, 2026](#)). Australian researchers propose a resale certification with simple grades to build buyer confidence ([UniSA, 2025](#)); any such scheme needs a cheap, repeatable power measurement.

■ Users and context

User	Need	Context
Second-life panel refurbisher or collector	Grade each module (A, B, C or reject) against its nameplate and record the result	Yard or warehouse, dozens to hundreds of modules a day, outdoor sun on a rack
Small installer or solar technician	Find why a rooftop or off-grid array underperforms: shading, a bad module, a hot connector	Rooftops and ground mounts, single modules or short strings, often in hot sun
Buyer of used panels (NGO, cooperative, household)	Check that a used module delivers what the seller claims before paying	Market stall or depot; little training
Trainer or TVET college	Teach how an IV curve reveals faults, with an instrument students can build	Classroom and outdoor bench
Researcher	Log curves over months for degradation studies at low cost	Outdoor test rack

Typical modules under test are 36-cell (about 22 V open circuit) to 144 half-cell (about 50 V, 12 to 14 A short circuit). Two older modules in series reach about 75 V.

■ Prior work

- **IV Swinger 2** by Chris Satterlee is an open-source tracer of about \$50 in parts. It uses load capacitors switched by relays or SSRs and drained through a bleed resistor, and runs from a laptop over USB ([GitHub](#); [firmware comments](#)). PVTrace keeps the capacitive load and adds standalone field use, irradiance and temperature sensing, STC translation and grading.
- **Cáceres et al. (2020)** built a capacitive-load tracer for 0 to 100 V and 0 to 12 A for under 200 EUR in materials, aimed at training and research centers that cannot afford commercial tracers ([Energies](#)).
- **González et al. (2021)** describe an open IoT tracer on a Raspberry Pi with a programmable electronic load, noting that commercial systems are "generally expensive and closed for modifications" ([Sensors](#)).
- **Commercial tracers** cover strings to 1,000 V or more and are priced for professional crews ([Transcat listing](#)). Industry guidance is that curves converted to standard test conditions (STC) should be taken at 700 W/m² or more, per IEC 61829 ([Seaward](#)).

The gap PVTrace addresses is a handheld, battery-powered, open tracer that works without a laptop, measures irradiance and module temperature, translates to STC and applies a transparent grading rule.

■ Constraints

- Garage-buildable prototype, about \$150 USD in parts (`project.yaml` budget).
- Off-the-shelf parts and hand tools; no custom machining.
- Voltage kept below 120 V DC, the extra-low-voltage limit for ripple-free DC in IEC 61140 ([summary](#)). Full strings at 600 to 1,500 V are out of scope.
- Works in full sun at up to 45 °C ambient, with no mains power.
- Open data: every sweep saved as a plain CSV that anyone can reprocess.

■ Out of scope

- Strings above 100 V open circuit and any work on live grid-connected arrays without isolation by the system owner.
- Insulation resistance, earth continuity and wet leakage tests (IEC 62446-1 commissioning tests beyond the IV curve).
- Electroluminescence or thermal imaging, which show cracks directly; PVTrace infers them from the curve only.
- Certifying modules for sale; PVTrace grades are indicative until a scheme owner adopts them.

■ Co-design checklist

- [] Identify a second-life panel partner (refurbisher, recycler or reuse program) to agree grade thresholds and throughput.
- [] Interview two or more small installers on which faults they meet most and how they find them today.
- [] Confirm the module types in the partner's stock (cell count, Voc, Isc) to fix the voltage and current ranges.
- [] Agree what a buyer needs to see on a printed or phone grade label.
- [] Check local rules for handling and reselling used modules (for example, WEEE in the EU).

■ Open questions

- Which grade thresholds would a buyer or scheme trust, and must they match an emerging standard?
- Is a reference cell calibrated once against a pyranometer good enough, or does each unit need its own calibration?
- How many sweeps per module make a reliable grade under passing clouds?

Safety: PV modules produce voltage whenever light falls on them and cannot be switched off. Short-circuit current flows as soon as the terminals are bridged, and a DC arc does not self-extinguish like an AC arc. Never break a DC circuit under load, and never connect the tracer to a string above its rating.